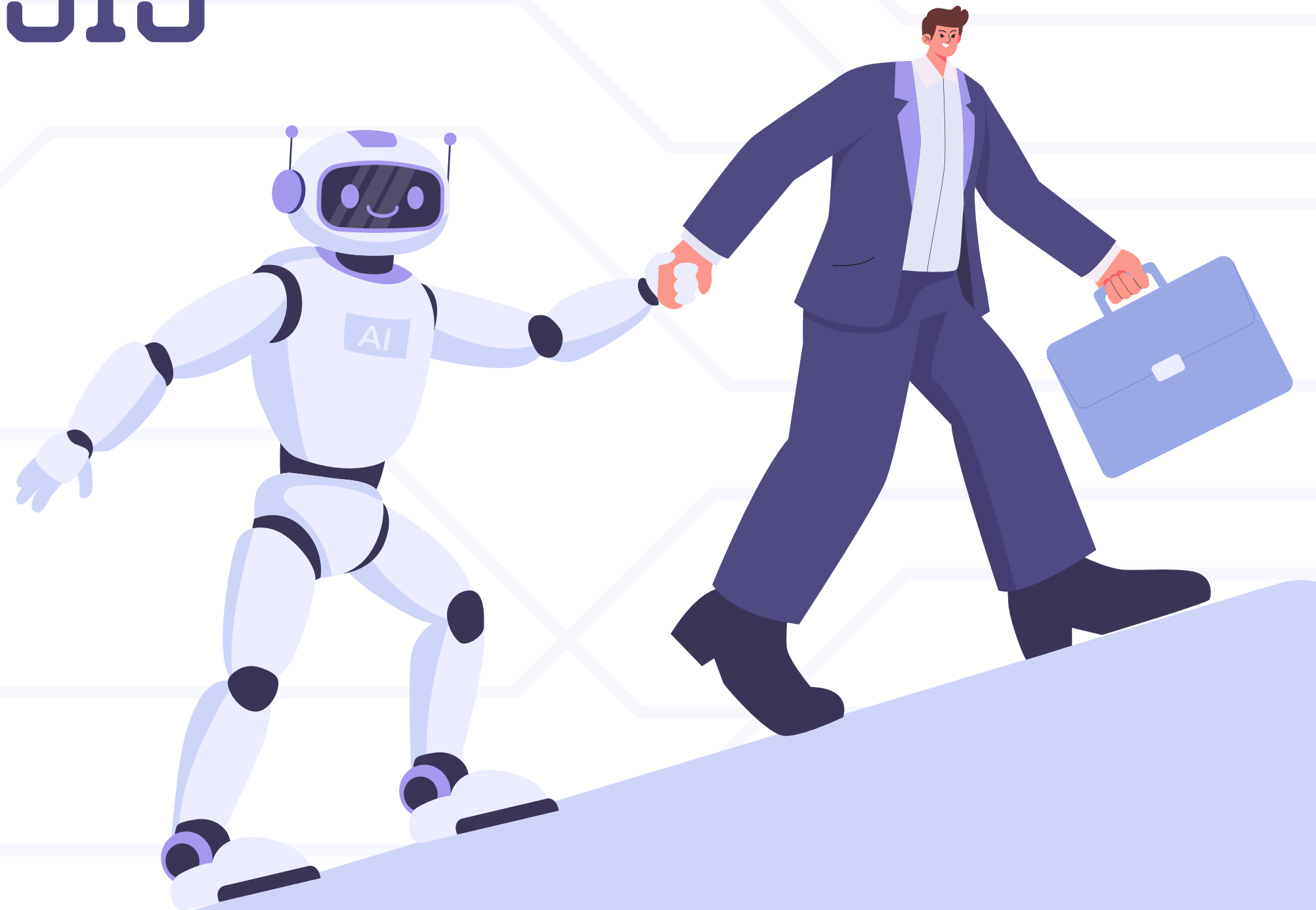


PREDICTIVE TASK DURATION ANALYSIS



**PRESENTED BY:
ABANOUB BARAYO**

✿ PROBLEM STATEMENT

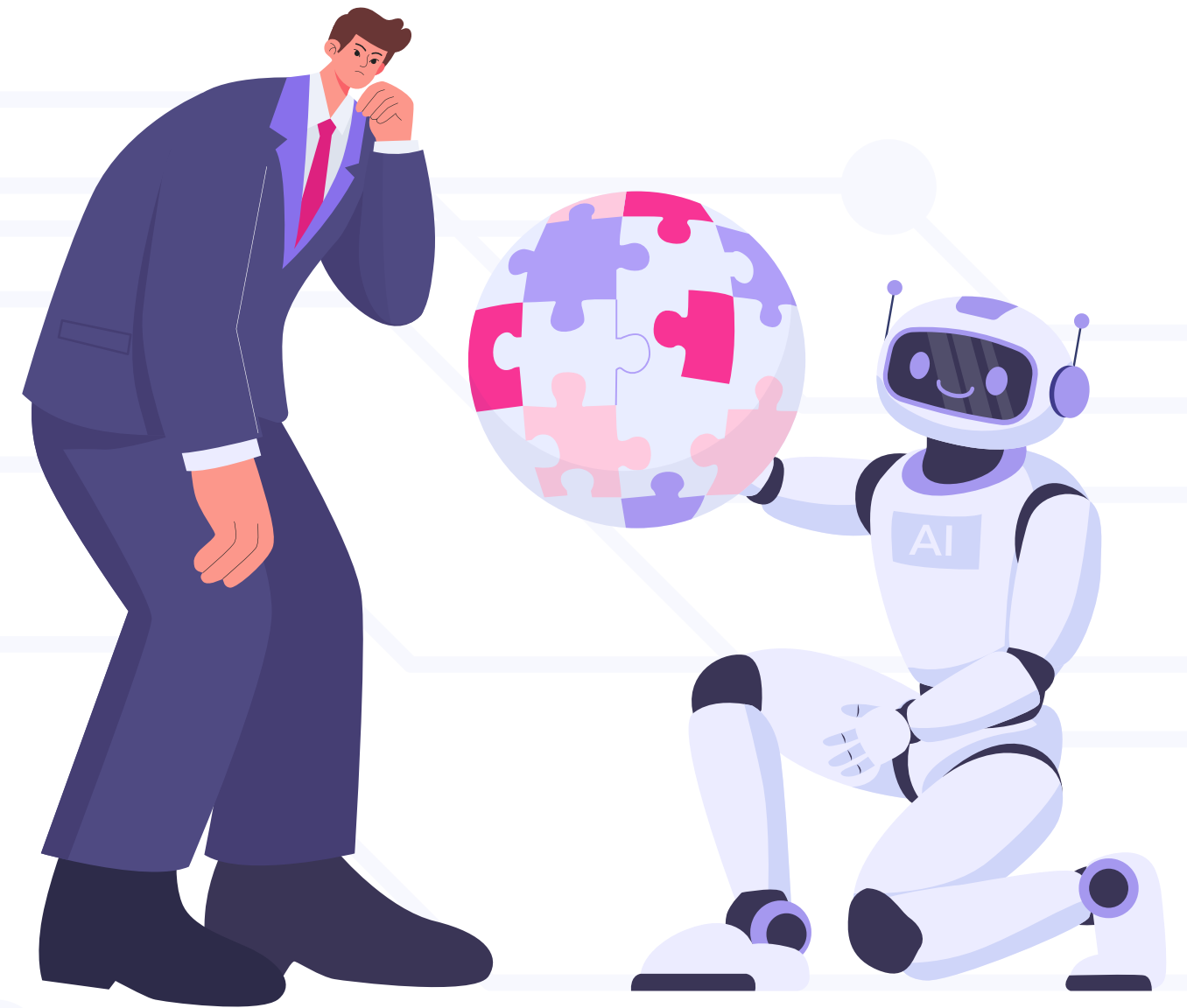
Software development estimation is often subjective, leading to "planning fallacies" and missed deadlines.

✿ OBJECTIVE

To transition from intuitive estimation to data-driven prediction of task completion times.

✿ RESEARCH QUESTION

Can we accurately forecast Task_Duration_Hours using a combination of technical, workflow, and human-centric data?



TECHNICAL EXECUTION – DATA HANDLING

	Hours_Coding	Lines_of_Code	Bugs_Found	AI_Usage_Hours	Sleep_Hours	Cognitive_Load	Coffee_Intake	Stress_Level	Commits
0	7	416	9	6	5.9	92	7	99	20
1	4	269	16	5	5.1	85	2	100	17
2	11	439	3	2	6.2	38	2	55	35
3	8	472	15	4	4.2	26	5	30	28
4	5	265	19	5	8.1	82	6	82	25
...	...	...	...	...	...	...	...	...	...
995	10	660	14	0	5.5	21	2	30	20
996	9	484	13	1	8.8	34	0	39	37
997	2	128	4	3	8.7	42	3	49	5
998	8	266	5	1	5.7	48	0	63	9
999	8	604	16	3	6.9	26	4	30	11



MODEL INPUTS

- Activity: Coding hours, lines of code, and bugs found.
- Tools: Total AI usage hours.
- Context: Sleep duration, cognitive load, and caffeine intake.



STANDARDIZATION

- Use StandardScaler
- Normalized all features to a common scale.
- Prevents high-value features (Lines of Code) from mathematically dominating low-value features (Coffee Intake).



VALIDATION STRATEGY

- 80/20 Split: 80% for model training; 20% reserved for testing.
- Ensures the model is evaluated on "unseen" data to verify predictive reliability.

LINEAR REGRESSION



BASELINE STRATEGY

Initiated analysis with Linear Regression to establish a performance benchmark.



TECHNICAL IMPLEMENTATION

Utilized OLS method to minimize the sum of squared differences between observed and predicted values



PERFORMANCE METRICS

MAE: 3.00 hours , RMSE: 3.93 hours ,R2: 57%



KEY DRIVERS OF TASK DURATION

THE MAJOR TIME DRIVER

Hours Coding (+4.64): Every hour of active coding is the strongest predictor of total task length.

EFFICIENCY BOOSTERS (TIME SAVERS)

Cognitive Load (-0.28) & Sleep (-0.17)
AI Usage (-0.14)

DURATION INCREASES (DELAY FACTORS)

Lines of Code (+0.24) & Stress (+0.22)
Coffee Intake (+0.21)

PRODUCTIVITY SIGNALS

Commits (-0.13) & Bugs Found (-0.13)



RANDOM FOREST MODELING

✿ THE ENSEMBLE APPROACH

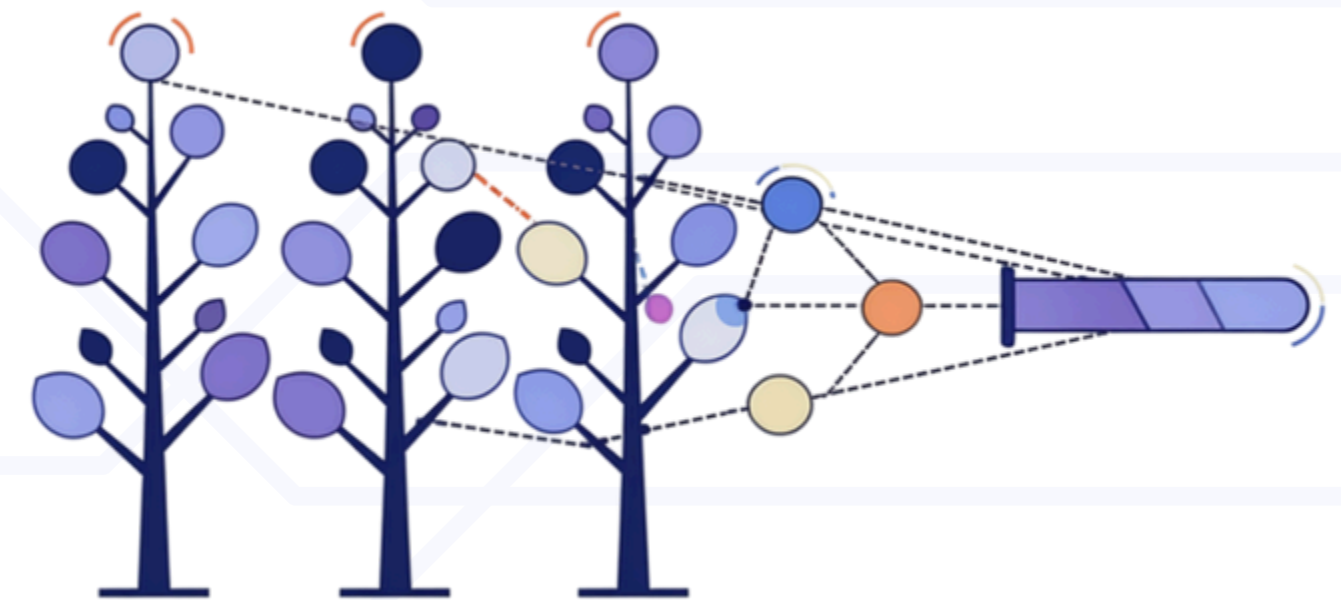
- Utilized a Random Forest Regressor to capture non-linear relationships.
- Built a "forest" of 100 individual decision trees (`n_estimators=100`) that collaborate to make a final prediction.

✿ PERFORMANCE EVALUATION

- R^2 : 53.37 %
- Mean Absolute Error (MAE): 3.13 hours.

✿ TECHNICAL INSIGHT

- The slight decrease in accuracy compared to Linear Regression (R^2 : 57%) suggests the underlying data patterns are predominantly linear rather than highly complex.



FEATURE IMPORTANCE – KEY DRIVERS



RANKING METHODOLOGY

- Used the feature_importances_ attribute to identify the most influential variables in the Random Forest model.



PRIMARY SIGNAL

- Hours_Coding is the dominant predictor of task duration.



SECONDARY DRIVERS

- AI_Usage_Hours and Stress_Level provide critical context for model accuracy.



ACTIONABLE INSIGHT

- Focus management efforts on these top-tier variables to improve project predictability.

REFINEMENT – FEATURE ENGINEERING & GBR



FEATURE ENGINEERING

- Created the AI-to-Manual Ratio and Total Active Hours to measure AI's relative efficiency.



OPTIMIZATION MODEL

- Implemented Gradient Boosting Regressor (GBR), which builds trees sequentially to minimize errors from previous stages.



OUTCOME

- R2: 51.86 %



TECHNICAL CONCLUSION

- Added complexity and custom features did not outperform the baseline, confirming that the primary relationships in this data are robustly linear.

MODEL COMPARISON



LINEAR REGRESSION

- MAE: 3.00h
- R2: 56.77 %



RANDOM FOREST

- MAE: 3.13h
- R2: 53.37 %



GRADIENT BOOSTING

- MAE: 3.17h
- R2: 51.86 %

CONCLUSION & RECOMMENDATIONS

- ✿ **LINEAR REGRESSION IS THE RECOMMENDED MODEL FOR THIS DATASET, OFFERING THE BEST BALANCE OF SIMPLICITY, TRANSPARENCY, AND PREDICTIVE ACCURACY**
- ✿ **AI USAGE HOURS CONSISTENTLY CORRELATE WITH REDUCED TASK DURATION, JUSTIFYING FURTHER INVESTMENT IN AI-ASSISTED WORKFLOWS.**
- ✿ **DEVELOPER WELL-BEING (SLEEP AND STRESS MANAGEMENT) IS A MEASURABLE FACTOR IN DELIVERY SPEED; THESE SHOULD BE INTEGRATED INTO CAPACITY PLANNING.**
- ✿ **PRIORITIZE MANAGING COGNITIVE LOAD AND STRESS LEVELS TO AVOID PROJECT BOTTLENECKS.**





THANK YOU!

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**GITHUB REPOSITORY
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DEVELOPER-PRODUCTIVITY-PERFORMANCE](https://github.com/abanooubdev/ironhack-ai-developer-productivity-performance)**